Use of AI Tools

Generative AI tools were used solely for language editing and grammatical correction during manuscript preparation. Specifically, a locally deployed large language model was used to improve grammar, phrasing, and readability. No generative AI system was used to develop the research ideas, formulate the methodology, conduct experiments, analyze or interpret results, generate scientific claims, or make substantive decisions regarding the content of the manuscript. All technical content, arguments, results, and conclusions were produced and verified by the authors, who take full responsibility for the final manuscript. Annotation was performed by human annotators as described in Section VI.

References

[1] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. tau Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[2] Z. Guo, M. Schlichtkrull, and A. Vlachos, "A survey on automated fact-checking," *Transactions of the Association for Computational Linguistics*, vol. 10, pp. 178–206, 2022.

[3] W. Chen, H. Zha, Z. Chen, W. Xiong, H. Wang, and W. Y. Wang, "HybridQA: A dataset of multi-hop question answering over tabular and textual data," *Findings of the Association for Computational Linguistics: EMNLP (Findings of EMNLP)*, pp. 1026–1036, 2020.

[4] J. Herzig, P. K. Nowak, T. Müller, F. Piccinno, and J. M. Eisenschlos, "TaPas: Weakly supervised table parsing via pre-training," in *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020, pp. 4320–4333.

[5] R. Aly, Z. Guo, M. Schlichtkrull, J. Thorne, A. Vlachos, C. Christodoulopoulos, O. Cocarascu, and A. Mittal, "FEVEROUS: Fact extraction and VERification over unstructured and structured information," in *NeurIPS Datasets and Benchmarks Track*, 2021.

[6] A. Narayan, I. Chami, L. Orr, and C. Ré, "Can foundation models wrangle your data?" in *Proceedings of the VLDB Endowment*, vol. 16, no. 4, 2022, pp. 738–746.

[7] R. C. Fernandez, A. J. Elmore, M. J. M. Franklin, S. Krishnan, and C. Tan, "How large language models will disrupt data management," in *Proceedings of the VLDB Endowment*, vol. 16, no. 11, 2023, pp. 3302–3309.

[8] T. J. Green, G. Karvounarakis, and V. Tannen, "Provenance semirings," in *ACM SIGACT-SIGMOD-SIGART Symposium on Principles of Database Systems (PODS)*, 2007, pp. 31–40.

[9] J. Cheney, L. Chiticariu, and W.-C. Tan, "Provenance in databases: Why, how, and where," *Foundations and Trends in Databases*, vol. 1, no. 4, pp. 379–474, 2009.

[10] R. F. Nogueira and K. Cho, "Passage re-ranking with BERT," *arXiv preprint arXiv:1901.04085*, 2019.

[11] O. Khattab and M. Zaharia, "ColBERT: Efficient and effective passage search via contextualized late interaction over BERT," in *International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*, 2020, pp. 39–48.

[12] H. Rashkin, V. A. Nikolaev, M. Lamm, L. M. Aroyo, M. Collins, D. Das, S. Petrov, G. S. Tomar, I. Turc, and D. Reitter, "Measuring attribution in natural language generation models," *Computational Linguistics*, vol. 49, no. 4, pp. 777–840, 2023.

[13] B. Bohnet, V. Q. Tran, P. Verga, R. Aharoni, D. Andor, L. B. Soares, M. Ciaramita, J. Eisenstein, K. Ganchev, J. Herzig, K. Hui, T. Kwiatkowski, J. Ma, J. Ni, L. S. Saralegui, T. Schuster, W. W. Cohen, M. Collins, D. Das, D. Metzler, S. Petrov, and K. Webster, "Attributed question answering: Evaluation and modeling for attributed large language models," *arXiv preprint arXiv:2212.08037*, 2022.

[14] N. F. Liu, T. Zhang, and P. Liang, "Evaluating verifiability in generative search engines," *Findings of the Association for Computational Linguistics: EMNLP (Findings of EMNLP)*, pp. 7001–7025, 2023.

[15] T. Gao, H. Yen, J. Yu, and D. Chen, "Enabling large language models to generate text with citations," in *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023, pp. 6465–6488.

[16] J. P. Thorne, A. Vlachos, C. Christodoulopoulos, and A. Mittal, "FEVER: a large-scale dataset for fact extraction and VERification," in *Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, 2018, pp. 809–819.

[17] Z. Yang, P. Qi, S. Zhang, Y. Bengio, W. W. Cohen, R. Salakhutdinov, and C. D. Manning, "HotpotQA: A dataset for diverse, explainable multi-hop question answering," in *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2018, pp. 2369–2380.

[18] L. Gao, Z. Dai, P. Pasupat, A. Chen, A. T. Chaganty, Y. Fan, V. Zhao, N. Lao, H. Lee, D.-C. Juan, and K. Guu, "RARR: Researching and revising what language models say, using language models," in *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023, pp. 16 477–16 508.

[19] E. Kamalloo, A. Jafari, X. Zhang, N. Thakur, and J. Lin, "HAGRID: A human-LLM collaborative dataset for generative information-seeking with attribution," *arXiv preprint arXiv:2307.16883*, 2023.

[20] C. J. C. Burges, T. Shaked, E. Renshaw, A. Lazier, M. Deeds, N. Hamilton, and G. Hullender, "Learning to rank using gradient descent," in *International Conference on Machine Learning*, 2005, pp. 89–96.

[21] F. Xia, T. Liu, J. Wang, W. Zhang, and H. Li, "Listwise approach to learning to rank," in *International Conference on Machine Learning*, 2008, pp. 1192–1199.

[22] V. Karpukhin, B. Oğuz, S. Min, P. Lewis, L. Wu,

S. Edunov, D. Chen, and W. tau Yih, "Dense passage retrieval for open-domain question answering," in *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020, pp. 6769–6781.

[23] S. E. Robertson and H. Zaragoza, "The probabilistic relevance framework: BM25 and beyond," *Foundations and Trends in Information Retrieval*, vol. 4, no. 1-2, pp. 1–174, 2009.

[24] Z. Wang, P. Ng, X. Ma, R. M. Nallapati, and B. Xiang, "Multi-passage BERT: A globally normalized BERT model for open-domain question answering," in *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2019, pp. 5877–5881.

[25] M. Freitag and Y. Al-Onaizan, "Beam search strategies for neural machine translation," in *Workshop on Neural Machine Translation (WNMT)*, 2017, pp. 56–60.

[26] C. Chow, "On optimum recognition error and reject tradeoff," *IEEE Transactions on Information Theory*, vol. 16, no. 1, pp. 41–46, 1970.

[27] Y. Geifman and R. El-Yaniv, "Selective classification for deep neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 4878–4887.

[28] A. Kamath, R. Jia, and P. Liang, "Selective question answering under domain shift," in *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020, pp. 5684–5696.

[29] J. Xin, R. Tang, Y. Yu, and J. Lin, "The art of abstention: Selective prediction and error regularization for natural language processing," in *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2021, pp. 1040–1051.

[30] A. N. Angelopoulos and S. Bates, "Conformal prediction: A gentle introduction," *Foundations and Trends in Machine Learning*, vol. 16, no. 4, pp. 494–591, 2023.

[31] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster, "Conformal risk control," in *International Conference on Learning Representations (ICLR)*, 2024.

[32] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *International Conference on Machine Learning (ICML)*, 2017, pp. 1321–1330.

[33] J. DeYoung, S. Jain, N. F. Rajani, E. Lehman, C. Xiong, R. Socher, and B. Wallace, "ERASER: A benchmark to evaluate rationalized NLP models," in *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020, pp. 4443–4458.

[34] R. Reiter, "A theory of diagnosis from first principles," *Artificial Intelligence*, vol. 32, no. 1, pp. 57–95, 1987.

[35] A. Ignatiev, N. Narodytska, and J. P. Marques-Silva, "Abduction-based explanations for machine learning models," in *AAAI Conference on Artificial Intelligence*, 2019.

[36] P. Buneman, S. Khanna, and W.-C. Tan, "Why and where: A characterization of data provenance," in *International Conference on Database Theory (ICDT)*, 2001, pp. 316–330.

[37] A. Meliou, W. Gatterbauer, K. F. Moore, and D. M. Suciu, "The complexity of causality and responsibility for query answers and non-answers," in *Proceedings of the VLDB Endowment*, vol. 4, no. 1, 2010, pp. 34–45.

[38] W. Chen, H. Wang, J. Chen, Y. Zhang, H. Wang, S. Li, X. Zhou, and W. Y. Wang, "TabFact: A large-scale dataset for table-based fact verification," in *International Conference on Learning Representations (ICLR)*, 2020.

[39] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang, "Lost in the middle: How language models use long contexts," *Transactions of the Association for Computational Linguistics*, vol. 12, pp. 157–173, 2024.

[40] W. Kryściński, B. McCann, C. Xiong, and R. Socher, "Evaluating the factual consistency of abstractive text summarization," in *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020, pp. 9332–9346.

[41] P. Laban, T. Schnabel, P. N. Bennett, and M. A. Hearst, "SummaC : Re-visiting NLI-based models for inconsistency detection in summarization," *Transactions of the Association for Computational Linguistics*, vol. 10, pp. 163–177, 2022.

[42] Y. Zha, Y. Yang, R. Li, and Z. Hu, "AlignScore: Evaluating factual consistency with a unified alignment function," in *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023, pp. 11 328–11 348.

[43] F. Petroni, A. Piktus, A. Fan, P. Lewis, M. Yazdani, N. D. Cao, J. P. Thorne, Y. Jernite, V. Karpukhin, J. Maillard, V. Plachouras, T. Rocktäschel, and S. Riedel, "KILT: a benchmark for knowledge intensive language tasks," in *Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, 2021, pp. 2523–2544.

[44] N. Thakur, N. Reimers, A. Rücklé, A. Srivastava, and I. Gurevych, "BEIR: A heterogenous benchmark for zero-shot evaluation of information retrieval models," in *NeurIPS Datasets and Benchmarks Track*, 2021.

[45] S. Chiesurin, D. Dimakopoulos, M. A. S. Cabezudo, A. Eshghi, I. V. Papaioannou, V. Rieser, and I. Konstas, "The dangers of trusting stochastic parrots: Faithfulness and trust in open-domain conversational question answering," *Findings of the Association for Computational Linguistics: ACL (Findings of ACL)*, pp. 947–959, 2023.

[46] P. He, J. Gao, and W. Chen, "DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing," in *International Conference on Learning Representations (ICLR)*, 2023.

[47] R. Artstein and M. Poesio, "Inter-coder agreement for computational linguistics," *Computational Linguistics*, vol. 34, no. 4, pp. 555–596, 2008.

APPENDIX A
PROOF OF PROPOSITION 1

We prove the statement for the benchmark pool construction that Section VI instantiates in the Key-Switch diagnostic. Let the claim c require one text value and one table value that agree on a key. The pool contains text units a_1, a_2 and table units b_1, b_2. Assume an additive scorer $f(c, S) = \sum_{u \in S} h(c, u) + \beta$ with a single threshold τ, exact constraints, and no ties. Suppose, as the benchmark asserts, that $\{a_1, b_1\}$ and $\{a_2, b_2\}$ are minimally sufficient, so each is sufficient and neither is redundant, and that the cross-paired sets $\{a_1, b_2\}$ and $\{a_2, b_1\}$ are insufficient because their keys mismatch. Sufficiency and insufficiency of sets of size two give four strict constraints:

$$h(a_1) + h(b_1) + \beta \geq \tau, \quad h(a_2) + h(b_2) + \beta \geq \tau, \quad (12)$$

$$h(a_1) + h(b_2) + \beta < \tau, \quad h(a_2) + h(b_1) + \beta < \tau. \quad (13)$$

Adding the two sufficiency inequalities yields $h(a_1)+h(b_1)+h(a_2)+h(b_2)+2\beta \geq 2\tau$. Adding the two insufficiency inequalities yields the same left-hand side strictly below 2τ. The two cannot both hold, so no additive scorer with one threshold classifies all four sets correctly. No step uses the values of h or the sign of β, so the conclusion holds for every additive scorer, and it is a representational obstruction rather than a fitting problem: the additive class assigns the same total to a pairing and to its cross-pairing whenever the multisets of unit scores coincide, and no choice of threshold can separate them.

For the positive direction it suffices to exhibit a scorer in the set-conditioned class that classifies the pool correctly, and to note that the pattern it implements is computable from the concatenation. Define $g(c, S) = \tau+1$ if S contains at least one text unit and at least one table unit, and every such pair agrees on the key required by c; and $g(c, S) = \tau - 1$ otherwise. The predicate is a function of the joint token sequence of c and S: it compares the key field of a text unit with the header-derived key of a cell, which is exactly the comparison that the cross-attention of Section IV can compute, since tokens of a_1 attend to the headers of b_2 inside one sequence. The two guards matter: without them the predicate "every text unit and every table unit agree on the key" is vacuously true for a singleton, which would classify the size-one sets as sufficient and violate the requirement that size-one sets be classified. With them, singletons and homogeneous pairs score $\tau - 1$ and are classified insufficient, so g separates $\{a_1, b_1\}, \{a_2, b_2\}$ from $\{a_1, b_2\}, \{a_2, b_1\}$ and from all size-one sets under the same threshold τ. The proposition asserts existence of such a scorer, not that gradient training finds it; whether the trained model realises the separation is the empirical question that Section VII-A answers on 300 locked Key-Switch instances.

APPENDIX B
INFERENCE AND PERMUTATION PROTOCOL

Algorithm 2 states the deployment policy of Section III. If $n = 0$ it returns $\perp$ with $r_\theta = -\infty$; otherwise it scores all singletons, keeps the top B, runs the beam with canonical ordering and deduplication, computes r_θ by (3) and abstains if $r_\theta < \tau$.

Algorithm 2 CertSet inference

input: c, pool $\mathcal{P}$ with $|\mathcal{P}| = n$, θ, τ, B, b, K
if $n = 0$ **then**
  return $\perp$ with $r_\theta = -\infty$
end if
score all n singletons; keep the top B into $\mathcal{B}_\theta$
$\Omega \leftarrow \{\{u\} : u \in \mathcal{B}_\theta\}$
for $k = 2$ **to** K **do**
  extend each retained set of size $k-1$ with each shortlisted unit not in it
  deduplicate candidates by index tuple hash
  score candidates; retain the top b into $\Omega^{(k)}$
end for
$M \leftarrow \max_{S \in \Omega} f_\theta(c, S)$; $\rho \leftarrow B^{-1} \sum_{u \in \mathcal{B}_\theta} f_\theta(c, \{u\})$; $r_\theta \leftarrow M - \rho$
if $r_\theta < \tau$ **then**
  return $\perp$
else
  return $\arg\max_{S \in \Omega} f_\theta(c, S)$ {ties broken lexicographically}
end if

Permutation protocol: Each development and test claim is presented 20 times per checkpoint with a pseudo-random permutation of its selected units, drawn from a fixed random stream (identifier 41). For each permutation we record $f_\theta(c, S)$ for the canonical-order argmax and the argmax under the permuted ordering. Score SD is the standard deviation of f_θ across permutations for the same set; decision instability is the fraction of claims whose beam argmax changes under at least one of the 20 permutations. In the slot-embedding ablation the slot embeddings are replaced by a shared zero vector, which preserves parameter count, and in the header ablation header tokens are replaced by a mask token rather than removed, which preserves sequence length.

APPENDIX C
ANNOTATION AND ADJUDICATION PROTOCOL

Two annotators read the claim and the full pool in randomised order with no model output and independently enumerate $\mathcal{M}^{\text{obs}}$ and verified negatives; a third adjudicates disagreements. A set supports the claim if and only if every entity, quantity and relation in the claim is grounded in the set, the asserted relation is stated in or directly entailed by the union of the units, and the inference uses nothing beyond common definitions and arithmetic on the cited cells. A pool is contradicted, and labelled $z = 0$, if some unit entails the negation of the claim under those conditions. Jointly cited units must agree on entity, measurement unit and period, with table headers read as the key. A set is redundant if it supports the claim and removing some unit still supports it, and minimal if it supports the claim and no proper subset does; minimality is checked by deletion, with at most K judgments per set. To

enumerate, each annotator marks the units that are individually relevant and then enumerates minimal sets of size at most K exhaustively within that subset, stopping when a full sweep adds nothing new; $\mathcal{M}^{\text{obs}}$ is the union over the two annotators.

For the audit, the expert applies a high-sensitivity relevance filter that keeps every unit mentioning any entity, quantity or relation of the claim, or a bridge entity linked to one, and then adjudicates *every* set of size at most four inside the filtered units, which is at most 385 sets at the ten-unit cap. Any output of any evaluated system that the expert judges sufficient is added to the audited set. A pool is in the audited positive family if this process yields a minimal sufficient set. Pools whose filtered set exceeds ten units are counted and reported as unresolved (38 pools, 7.9% of the sample) rather than silently dropped. The audit estimand is therefore exact under assumption (F) and interval-identified otherwise, which is why the paper reports a bound as well as a point estimate.

APPENDIX D
ADDITIONAL RESULTS

A. Secondary metrics

Table V extends Table I to the secondary endpoints. CertSet is first among the trained systems on every column: set-level recovery (68.4% against 61.8–65.6%), gold-unit hit rate (81.0% against 75.2–79.2%), support precision (87.2% against 81.1–84.1%), necessity (64.8% against 57.4–63.1%) and Key-Switch cross-key error (11.7% against 17.7–23.3%), and it combines the highest coverage with the lowest risk. The exhaustive-search control is above CertSet on each column by at most 1.4 points, the same pattern as on the primary endpoint. The RARR adaptation, whose original purpose is to revise generated text rather than to select citations, is last on every column.

TABLE V
SECONDARY METRICS ON THE FEVER LOCKED TEST, IN PERCENT. RISK IS AMONG NON-ABSTAINING OUTPUTS AND COVERAGE IS NON-ABSTAINING COVERAGE; CROSS-KEY ERROR IS MEASURED ON THE 300 KEY-SWITCH TEST INSTANCES. INTERVALS FOR CERTSET: SET-LEVEL RECOVERY 66.8–70.0, SUPPORT PRECISION 85.7–88.6, NECESSITY 62.7–66.8, RISK 11.4–14.3, COVERAGE 68.5–71.7.

Method	Set recov.	Gold hit	Support prec.	Neces-sity	Cross-key error	Risk / cov.
CertSet	68.4	81.0	87.2	64.8	11.7	12.8 / 70.1
Additive	63.7	77.0	81.9	60.1	21.3	16.6 / 68.8
Singleton+pair	65.6	79.2	84.1	61.9	17.7	15.0 / 69.4
RankNet	65.1	78.6	83.6	62.6	18.3	15.2 / 69.1
ListMLE	64.8	78.3	83.3	63.1	18.7	15.4 / 69.0
ALCE adapt.	62.9	76.4	82.4	58.8	22.3	16.4 / 68.1
RARR adapt.	61.8	75.2	81.1	57.4	23.3	17.2 / 67.4
Exhaustive	69.1	81.8	87.9	65.9	10.3	11.8 / 70.6

B. Calibration across domains

Figure 4 plots the AURC rows of Table II. In every column the full objective has the lowest AURC, the variant without the affine map is intermediate, and the variant without the calibration term is highest. Transfer raises all three objectives by similar amounts (0.038 to 0.061), which is why their order survives even though no transferred value matches the in-domain one.

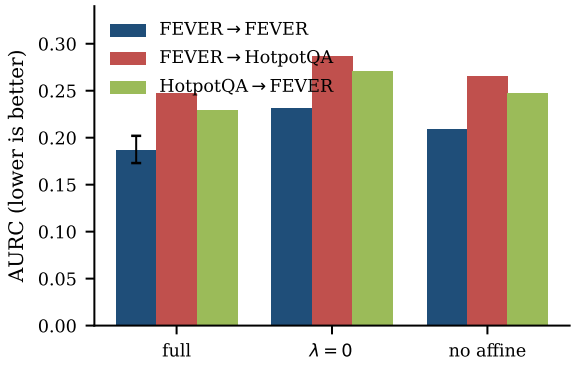


Fig. 4. AURC by objective and evaluation direction (lower is better). The full objective is best in every direction, the variant without the affine map is intermediate, and the variant without the calibration term ($\lambda = 0$) is worst. The error bar is the in-domain 95% interval for the full objective.

C. Negative strata

Table VI reports the stratum-removal experiment of Section VII-D. The construction reached 11,867 of 12,000 targeted held-out negatives, balanced across domain and stratum, with every held-out claim, pool, candidate set and source document disjoint from training. Even with one stratum removed, each variant falsely accepts fewer held-out negatives (12.2–13.9%) than the RankNet and ListMLE baselines (15.1% and 14.6%), so the verified-negative protocol improves held-out validity as a whole. On the locked test, removing any stratum lowers support precision and necessity and raises false acceptance on $z = 0$ pools (from 4.3% to 4.8–5.6%); the substitution stratum matters most on all four locked-test columns.

TABLE VI
STRATUM REMOVAL. HELD-OUT FALSE ACCEPTANCE (FA) IS MEASURED ON 11,867 ADJUDICATED NEGATIVES DISJOINT FROM TRAINING; THE REMAINING COLUMNS ARE ON THE 3,000-CLAIM FEVER LOCKED TEST, WITH FA ON $z = 0$ POOLS. p-VALUES ARE PAIRED MCNEMAR ON LOCKED-TEST RECOVERY WITH BH CORRECTION; THE HELD-OUT CONTRAST MARKED † USES A MCNEMAR TEST ON THE HELD-OUT NEGATIVES. THE RANKNET AND LISTMLE ROWS ARE TRAINED BASELINES EVALUATED ON THE HELD-OUT NEGATIVES ONLY.

Training negatives	Held-out	Locked test				
	FA	Rec.	Sup.	Nec.	FA	p
Full $O+U+H$	11.8	61.1	87.2	64.8	4.3	–
minus O	13.1	59.8	86.1	63.1	5.1	0.0128
minus U	12.2	59.2	85.7	62.6	5.6	1.7×10^{-4}
minus H	13.9†	60.4	86.6	64.0	4.8	0.083
RankNet	15.1	–	–	–	–	–
ListMLE	14.6	–	–	–	–	–

D. Ablation ledger

Table VII lists every ablation of Section VII-D and Figure 5 plots both ladders. Among the subtractive rows, the structural components (unit order, context headers, slot embeddings) cost

the most recovery and also raise decision instability, from 6.7% for the full model to 11.9–16.3%; score SD across permutations rises from 4.1 to 7.2 without headers and 8.9 under random order. Removing the calibration term costs only 0.9 points of recovery but degrades probability quality most (Brier 0.211 and ECE 0.108 against 0.184 and 0.071). In the cumulative build-up every stage is significant ($p \leq 1.0 \times 10^{-4}$ except the last, $p = 0.028$), and Key-Switch accuracy rises from 42.1% at retriever top-1 to 68.9% once the cross-encoder is introduced, before any change to the negatives or calibration.

TABLE VII
ABLATION LEDGER ON THE FEVER LOCKED TEST (FIVE RUNS). SUBTRACTIVE ROWS REMOVE ONE COMPONENT FROM THE FULL MODEL (61.1%); BUILD-UP ROWS ADD COMPONENTS CUMULATIVELY. CO-ENDPOINTS ARE PERCENTAGES EXCEPT BRIER AND ECE; FA IS FALSE ACCEPTANCE ON $z = 0$ POOLS, INSTAB. IS DECISION INSTABILITY AND K-S IS KEY-SWITCH ACCURACY. p-VALUES ARE BH-CORRECTED WITHIN THE FAMILY.

Condition	Recovery	Co-endpoints	BH p
Subtractive			
− omission stratum O	59.8	support 86.1; FA 5.1	0.0128
− substitution stratum U	59.2	support 85.7; FA 5.6	1.7×10^{-4}
− model-hard stratum H	60.4	support 86.6; FA 4.8	0.083
− calibration term ($\lambda = 0$)	60.2	Brier 0.211; ECE 0.108	0.022
− context headers	58.9	set recov. 66.1; instab. 13.8	2.5×10^{-5}
slot embeddings → zero	59.4	set recov. 66.7; instab. 11.9	0.0017
random unit order	57.9	set recov. 64.8; instab. 16.3	7.0×10^{-7}
title masking	59.7	set recov. 67.2; instab. 10.2	0.0084
Cumulative build-up			
retriever top-1	31.4	set recov. 39.1; K-S 42.1	1.0×10^{-12}
+ additive scoring	52.1	set recov. 59.2; K-S 61.7	1.0×10^{-12}
+ cross-encoder	57.8	set recov. 65.1; K-S 68.9	5.0×10^{-28}
+ verified negatives	60.1	set recov. 67.4; FA 4.9	4.0×10^{-12}
+ calibration term	60.8	Brier 0.196; ECE 0.079	1.0×10^{-4}
+ affine map	61.1	Brier 0.184; ECE 0.071	0.028

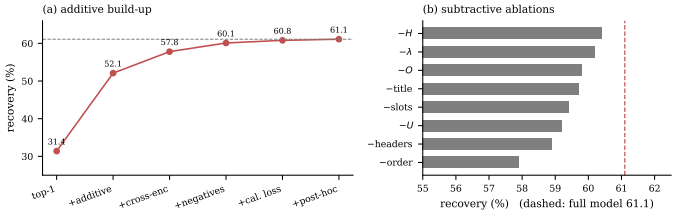


Fig. 5. (a) Cumulative build-up of recovery; (b) subtractive ablations against the full model's 61.1% (dashed). Unit order and context headers are the two largest single losses.

E. Hyperparameter grids

All grid points were selected on development data; Table VIII reports the locked-test recovery at each point, so it shows whether the development choices generalise. For the training hyperparameters they do: recovery peaks at the selected $\lambda = 0.3$ (60.2% at $\lambda = 0$, 59.8% at 0.1, 60.7% at 0.6 and 59.6% at 1.0) and at the selected margins $(1.0, 0.4)$, and the whole margin grid lies within a 2.4-point band, so the choice is not knife-edge. For search and capacity the largest setting is best: $B = 24$ beats $B = 12$ at both beam widths where both are reported, $(24, 32)$ is 0.3 points above the selected $(24, 16)$ at 65% more latency, and capacity is monotone but sublinear. The $(24, 8)$ recovery cell is omitted because its logged value could not be reconciled with the $b = 8$ beam-retention entry of Table III.

TABLE VIII
LOCKED-TEST RECOVERY (%) ACROSS THE HYPERPARAMETER GRIDS ON FEVER; SELECTED CELLS IN BOLD. MARGINS ARE $(m_{\mathsf{ins}}, m_{\mathsf{red}})$ WITH $m_{\mathsf{red}} < m_{\mathsf{ins}}$. THE $\lambda = 0$ ENDPOINT IS THE CALIBRATION-TERM ABLATION OF TABLE VII.

Grid	Setting	Recovery
λ	0.1 / 0.3 / 0.6 / 1.0	59.8 / **61.1** / 60.7 / 59.6
Margins	$(0.5, 0.2)$ / $(0.5, 0.4)$	58.7 / 59.1
	$(1.0, 0.2)$ / $(1.0, 0.4)$ / $(1.0, 0.8)$	59.9 / **61.1** / 60.3
	$(1.5, 0.2)$ / $(1.5, 0.4)$ / $(1.5, 0.8)$	59.6 / 60.7 / 59.8
(B, b)	$(12, 8)$ / $(12, 16)$ / $(12, 32)$	57.1 / 58.3 / 58.8
	$(24, 16)$ / $(24, 32)$	**61.1** / 61.4
Capacity	110M / 220M / 435M	56.2 / 58.9 / **61.1**

F. Frontier comparison

Table IX gives the per-model results behind Section VII-G. Every prompted model is below CertSet on both recovery (44.2–52.1% against 58.9%) and coverage (91.8–95.8% against 97.1%). Failures are small and counted rather than repaired: malformed outputs 0.4–1.4%, refusals 0.1–0.4% and timeouts 0.2–0.6%, all scored as abstentions. Latency is 2.3 to 6.3 times CertSet's. Per-domain results are reported for CertSet only (61.2% on FEVER, 57.0% on HotpotQA), because per-domain splits of the API budget were not planned in advance; the forced-answer run is a separate evaluation, and its CertSet figures are not directly comparable to the selective figures of Section VII. Two candidate successor models in the manifest were unavailable at access time; they were recorded with the provider error and excluded rather than substituted.

TABLE IX
FORCED-ANSWER COMPARISON ON THE COMBINED LOCKED TEST ($n = 6{,}703$). FAILURE COLUMNS ARE PERCENTAGES WITH COUNTS; EVERY FAILURE IS SCORED AS AN ABSTENTION. LATENCY (MS) IS END-TO-END FOR ONE CLAIM PER REQUEST, COLD CACHE. ROWS NAME MODEL FAMILIES; RESOLVED IDENTIFIERS, API VERSIONS AND ACCESS TIMESTAMPS ARE IN THE RELEASED MANIFEST.

Method	Rec.	Cov.	Malf.	Ref.	T/o	Lat.
CertSet	58.9	97.1	–	–	–	417
GPT-5 / o3 succ.	52.1	95.8	0.6 (40)	0.2 (13)	0.3 (20)	2,380
Claude Opus 4.5+	51.4	95.1	0.4 (27)	0.3 (20)	0.2 (13)	2,610
Gemini 2.5/3.0 succ.	50.2	94.6	0.7 (47)	0.1 (7)	0.4 (27)	2,190
DeepSeek-V3 / R1	48.7	93.9	0.9 (60)	0.2 (13)	0.5 (34)	1,870
Qwen3 largest	47.0	93.2	1.1 (74)	0.1 (7)	0.2 (13)	940
Llama 4 Maverick/Scout	44.2	91.8	1.4 (94)	0.4 (27)	0.6 (40)	1,120

APPENDIX E
IMPLEMENTATION, COST AND REPRODUCIBILITY

Implementation: The stack is PyTorch 2.5.1, Transformers 4.46.3 and CUDA 12.4 in bf16, with an effective batch size of 32, linear warmup over the first 10% of steps and linear decay. Training uses eight A100-80GB GPUs and inference is timed on one. Random streams control initialisation ordering, minibatch order, negative sampling and permutations; dataset

construction uses stream 17 and permutation diagnostics stream 41. Deterministic kernels are enabled where supported and nondeterministic operations are logged. Checkpoints are selected by development recovery in the target coverage interval, then by lower risk and earlier step.

Memory and compute: At the standard setting, peak allocated inference memory is 2,984 MB (95% CI 2,941–3,028): 1,740 MB of weights (equal to 435M parameters at four bytes each), 512 MB of activations, 582 MB of candidate tensors and 150 MB of serialised buffers; peak reserved memory is 3,310 MB. Gradients and optimiser states are absent because inference runs under `torch.no_grad`. Training the primary model consumed about 640 A100-80GB GPU-hours, and the 10,000-replicate prediction bootstrap needed 18.6 CPU-hours and no GPU.

Release: The release contains split manifests with claim IDs, component hashes and overlap counts; frozen pools with the retriever index hash; the transformation from evidence units to serialised strings; adjudication records for support, contradiction, redundancy and minimality; the audit sample, its filter output and the adjudication of every set of size at most four within it; the model manifest with resolved identifiers, timestamps and access records; and the code commit and checkpoint hash of the scorer. The adjudicators' identities and the API keys used for the frontier rows are not released. The nested bootstrap is released as a script that reproduces a reported interval from the recorded predictions.

APPENDIX F
SCOPE BOUNDARIES

Table X pairs each boundary of the evaluation with the quantity that measures it and the claim it rules out; Section VIII discusses the ones that matter most for deployment.

TABLE X
BOUNDARY CONDITIONS THE RESULTS MEASURE, AND THE SCOPE THEY IMPOSE.

Boundary	Measured quantity	Consequence
Exact search	62.3% exhaustive vs 61.1% beam (-1.2 pts; 95% CI -2.3 to -0.1; $p = 0.041$)	Recovery is conditional on the beam budget; exact search with the same scorer is better at 11.8× the latency
Retriever coverage	11.8% of gold units absent from the pool; oracle pool 69.2%	Absolute recovery is a property of this BM25+dense index
Cross-domain calibration	AURC 0.248 and 0.229 vs 0.187 in-domain	The absolute selective-prediction level is demonstrated in-domain only
Split construction	59.6% grouped vs 62.0% random; zero measured overlap on grouped splits	Headline figures come from grouped splits; random-split numbers are inflated
Audit filter	Filter sensitivity 91.8%; 151 units excluded, 14 in minimal sets; 38 pools unresolved; unresolved-as-positive 68.8% exceeds the 0.613 bound	The audit estimate is interval-identified under assumption (F), not point-identified
Timed axes	Only $n \in \{12, 100\}$ at 512 tokens timed	No latency or memory claim outside the measured axes
Unit types	Table cells exercised by Key-Switch and serialisation controls only	The text-plus-table claim is not evaluated on a table corpus with gold cells
Admissible witness	Cap $K = 4$; size distribution of $\mathcal{M}$ not measured	Aggregate, join and multi-row claims whose only witness exceeds four units are out of scope; the censored fraction is unknown
Comparators not run	No rule-based key matcher, no trained verifier, no coverage-matched frontier point	The frontier result is a prompted-model comparison only; a deterministic matcher may be near-exact on Key-Switch
Labels	No inter-annotator agreement, annotation volume or cost	Label noise is unquantified and the labelling cost of adopting the protocol is unpriced
Pool trust	Pools are adversarially clean by construction	The abstain action is a coverage control, not a defence against poisoned retrieval